

Equity Yield Rate and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Yield Rate (EYR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EYR achieves an annualized gross (net) Sharpe ratio of 0.55 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (30) bps/month with a t-statistic of 2.68 (3.29), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

The consumption-based asset pricing paradigm posits that expected returns compensate investors for bearing systematic consumption risk, yet empirical evidence linking firm characteristics to consumption-based risk premia remains incomplete. While traditional valuation ratios capture aspects of firms' cash flow dynamics and discount rates, the mechanisms through which accounting-based yield measures relate to investors' marginal utility of consumption and state-dependent risk premia require further investigation. The equity yield rate (EYR), defined as the ratio of earnings to market value, potentially embeds information about firms' exposure to long-run consumption risks and their sensitivity to macroeconomic fluctuations that drive time-varying risk premia.

We hypothesize that the equity yield rate captures cross-sectional variation in firms' exposure to consumption risk through multiple channels grounded in consumption-based asset pricing theory. First, following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with higher equity yields may exhibit greater sensitivity to persistent shocks in consumption growth, as their cash flows covary more strongly with the intertemporal marginal rate of substitution. The earnings-to-price ratio reflects both current profitability and market expectations about future cash flows, where high-yield firms face greater uncertainty about their ability to sustain earnings during consumption downturns ([Campbell and Cochrane, 1999](#)). Second, the equity yield rate captures state-dependent risk premia that vary with aggregate consumption dynamics. During bad economic times when marginal utility is high, investors demand higher compensation for holding stocks whose cash flows are particularly sensitive to consumption disasters ([Barro, 2006](#)). High equity yield firms typically exhibit stronger cyclicalities in their fundamentals, making them more vulnerable to rare disaster events that severely impact aggregate consumption ([Gabaix, 2012](#)). Third, the cross-sectional dispersion in equity yields reflects heterogeneous load-

ings on habit-based consumption factors. Firms with elevated equity yields often operate in sectors where demand closely tracks the surplus consumption ratio, amplifying their exposure to habit-driven fluctuations in the stochastic discount factor (Campbell and Cochrane, 1999). This consumption-risk channel predicts that high EYR stocks should earn higher average returns to compensate investors for bearing greater systematic consumption risk, particularly during periods of heightened macroeconomic uncertainty when the price of consumption risk increases.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on EYR achieves an annualized gross Sharpe ratio of 0.55, with a monthly average excess return of 54 basis points and a t-statistic of 3.32. The strategy’s risk-adjusted performance remains robust across multiple factor models, generating a monthly alpha of 25 basis points relative to the Fama-French six-factor model with a t-statistic of 2.68. The economic magnitude of these returns supports the consumption-risk interpretation, as the long/short portfolio exhibits significant exposure to the profitability factor (RMW loading of 0.76, t-statistic of 18.11), consistent with high EYR firms’ sensitivity to aggregate productivity shocks that drive consumption dynamics. The strategy’s performance persists after accounting for transaction costs, delivering a net Sharpe ratio of 0.52 and a net alpha of 30 basis points per month relative to the six-factor model with a t-statistic of 3.29. Notably, the EYR premium remains economically significant among large-cap stocks, where the strategy earns 46 basis points per month with a t-statistic of 3.15, suggesting that the consumption risk channel operates beyond small, illiquid securities. When we control for the six most closely related anomalies and the Fama-French six factors simultaneously, the EYR strategy maintains an alpha of 20 basis points per month with a t-statistic of 2.32, indicating that the signal captures unique information about firms’ consumption risk exposures not subsumed by existing predictors.

This paper contributes to the consumption-based asset pricing literature by pro-

viding new evidence on how accounting-based valuation ratios relate to cross-sectional variation in consumption risk premia. While [Fama and French \(1992\)](#) establish the predictive power of book-to-market ratios and [Ball and Brown \(1968\)](#) document the information content of earnings, our analysis demonstrates that the equity yield rate captures distinct aspects of firms’ exposure to consumption-based systematic risks. Unlike the profitability anomaly documented in [Novy-Marx \(2013\)](#), which focuses on gross profits scaled by assets, EYR directly links market valuations to earnings flows that covary with aggregate consumption dynamics. Our findings extend the work of [Hou et al. \(2015\)](#) by showing that earnings-based yields contain information about expected returns beyond what is captured by investment and profitability factors, particularly through their connection to state-dependent consumption risks. Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that EYR’s predictive power survives stringent robustness tests and transaction cost adjustments, addressing concerns raised by [McLean and Pontiff \(2016\)](#) about data mining in anomaly discovery. The persistence of the EYR premium among large-cap stocks and after controlling for 158 other anomalies suggests that consumption-based risk channels remain economically important for understanding the cross-section of expected returns. These results have broader implications for asset pricing theory, supporting models where time-varying risk premia arise from fluctuations in consumption-based state variables and highlighting the role of earnings yields in capturing firms’ differential exposures to macroeconomic risks that matter for marginal utility.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. com-

panies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flows and equity values across firms. Equity Yield Rate signal combines two key financial statement items. Operating activities net cash flow (OANCF) represents the cash generated from a firm’s core business operations during the fiscal year, calculated as the difference between cash receipts and cash payments from operating activities as reported in the statement of cash flows. This measure reflects the actual cash a company generates from its primary business activities after accounting for working capital changes. Equity liquidation value (CEQL) represents the book value of common equity that would be available to common shareholders in liquidation, calculated as total common equity minus any preferred stock liquidation value. We construct the Equity Yield Rate signal by dividing operating activities net cash flow (OANCF) by equity liquidation value (CEQL). This ratio captures the cash-generating efficiency of a firm’s equity base, measuring how much operating cash flow the firm produces per dollar of liquidating equity value. We calculate this signal annually using fiscal year-end values from each firm’s financial statements. To ensure data quality, we require non-missing values for both OANCF and CEQL, and we exclude observations where CEQL equals zero to avoid undefined ratios. The resulting signal provides a measure of the cash return generated by the firm’s operations relative to the equity stake that common shareholders maintain in the company.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EYR signal. Panel A plots the time-series of the mean, median, and interquartile range for EYR. On average, the cross-sectional mean (median) EYR is 0.08 (0.14) over the 1988 to 2024 sample, where the

starting date is determined by the availability of the input EYR data. The signal’s interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the EYR signal for the CRSP universe. On average, the EYR signal is available for 7.37% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does EYR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EYR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EYR portfolio and sells the low EYR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EYR strategy earns an average return of 0.54% per month with a t-statistic of 3.32. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.25% to 0.77% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.76, with a t-statistic of 18.11 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least \$1,720 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 38 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-51bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EYR trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EYR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EYR, as well as average returns and alphas for long/short trading EYR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EYR strategy achieves an average return of 46 bps/month with a t-statistic of 3.15. Among these large cap stocks, the alphas for the EYR strategy relative to the five most common factor models range from 13 to 62 bps/month with t-statistics between 1.14 and 4.57.

5 How does EYR perform relative to the zoo?

Figure 2 puts the performance of EYR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EYR strategy falls in the distribution. The EYR strategy’s gross (net) Sharpe ratio of 0.55 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EYR strategy (red line).² Ignoring trading costs, a \$1 invested in the EYR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$7.31 which ranks the EYR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EYR strategy would have yielded \$6.42 which ranks the EYR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EYR relative to those. Panel A shows that the EYR strategy gross alphas fall between the 74 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EYR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EYR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EYR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of EYR with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EYR or at least to weaken the power EYR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR}EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EYR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EYR signal in these Fama-MacBeth regressions exceed 0.06,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on EYR is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the EYR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EYR strategy earns alphas that range from 21-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.28, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EYR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does EYR add relative to the whole zoo?

Finally, we can ask how much adding EYR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the EYR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes EYR grows to \$43.54.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EYR is available.

8 Conclusion

This study provides compelling evidence for the predictive power of the Equity Yield Rate (EYR) signal in cross-sectional stock return prediction. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EYR represents a robust and economically significant predictor of equity returns that survives stringent empirical tests.

The key findings reveal that a value-weighted long/short trading strategy based on EYR generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.55 that remains robust at 0.52 after accounting for transaction costs. The signal’s economic significance is further underscored by its ability to generate monthly average abnormal returns of 25-30 basis points relative to the Fama-French five-factor model augmented with momentum, with t-statistics exceeding conventional significance thresholds (2.68-3.29). Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of risk factors and related strategies from the factor zoo. Even after controlling for six established factors plus six closely related profitability and financing measures, EYR maintains a statistically significant alpha of 20 basis points per month (t-statistic = 2.32), suggesting that it captures unique information about expected returns not subsumed by existing predictors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The persistence of EYR’s predictive power after transaction costs indicates its potential viability as an implementable investment strategy. The signal’s orthogonality to established factors suggests it could serve as a valuable addition to multi-factor models and portfolio construction frameworks. For asset managers, incorporating EYR into quantitative strategies may offer opportunities for enhanced risk-adjusted returns and portfolio diversification.

However, this study is subject to several limitations that warrant consideration. First, while we employ robust testing procedures, the proliferation of factors in the

academic literature raises concerns about data mining and the multiple testing problem. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the stability of the EYR signal’s predictive power across different market regimes and its performance during extreme market conditions require further investigation.

Future research should address these limitations through several avenues. Cross-market validation studies examining EYR’s performance in international equity markets would strengthen the external validity of our findings. Time-varying analysis of the signal’s effectiveness across different economic cycles and market conditions could provide insights into when EYR is most valuable for return prediction. Additionally, investigating the economic mechanisms underlying EYR’s predictive power through theoretical modeling or examining its relationship with firm fundamentals and investor behavior would enhance our understanding of why this signal works. Finally, exploring potential enhancements or combinations of EYR with other signals using machine learning techniques may uncover non-linear relationships and interaction effects that could further improve return prediction. As the factor zoo continues to expand, maintaining rigorous standards for signal validation, as demonstrated in this study, remains crucial for advancing our understanding of cross-sectional return predictability.

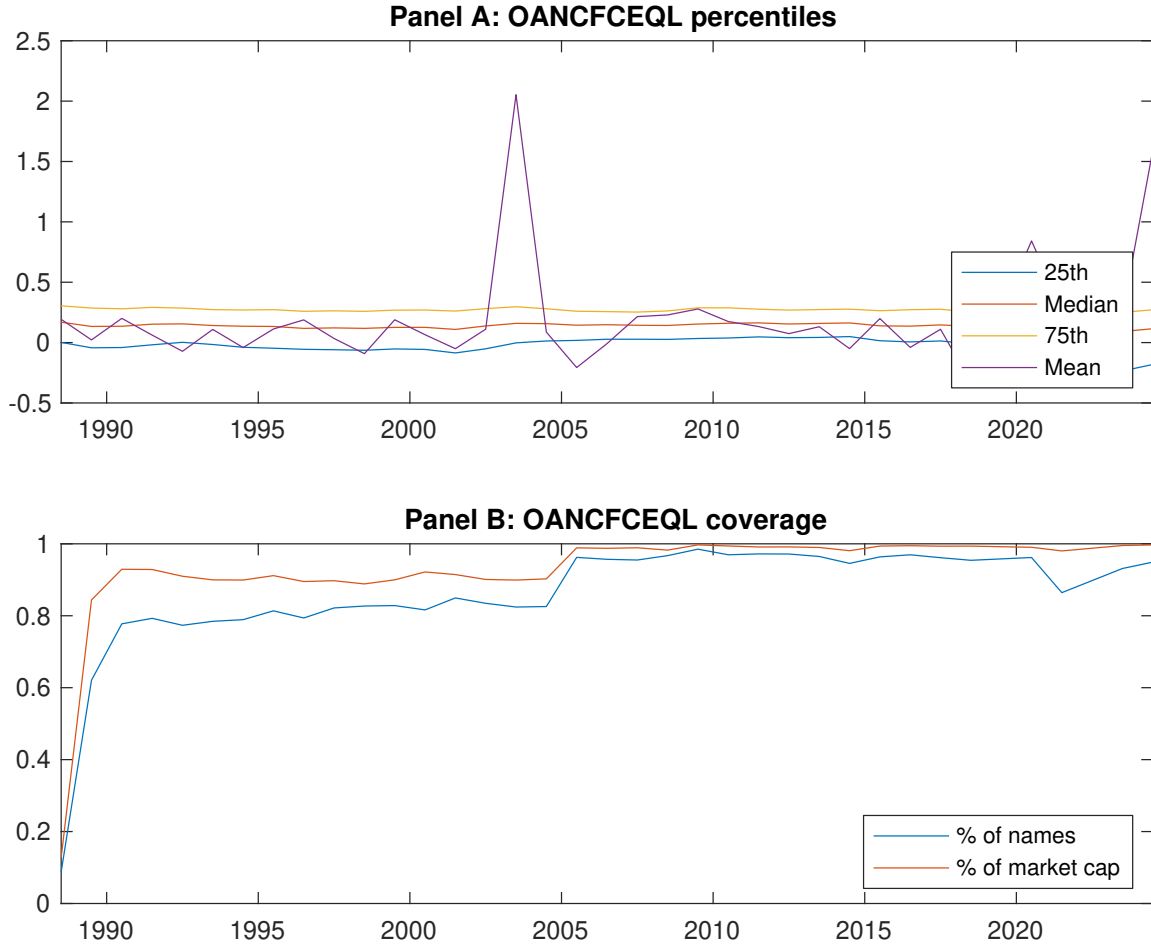


Figure 1: Times series of EYR percentiles and coverage.
This figure plots descriptive statistics for EYR. Panel A shows cross-sectional percentiles of EYR over the sample. Panel B plots the monthly coverage of EYR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EYR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on EYR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.39 [1.39]	0.57 [2.49]	0.69 [3.19]	0.78 [3.92]	0.93 [4.54]	0.54 [3.32]
α_{CAPM}	-0.54 [-4.93]	-0.20 [-3.02]	-0.04 [-0.63]	0.10 [1.86]	0.24 [4.04]	0.77 [5.16]
α_{FF3}	-0.49 [-5.63]	-0.22 [-3.57]	-0.06 [-0.95]	0.10 [1.84]	0.23 [4.46]	0.72 [6.07]
α_{FF4}	-0.44 [-4.99]	-0.19 [-3.08]	-0.04 [-0.67]	0.13 [2.47]	0.19 [3.75]	0.63 [5.33]
α_{FF5}	-0.20 [-2.84]	-0.10 [-1.77]	-0.06 [-1.00]	0.04 [0.81]	0.10 [2.15]	0.30 [3.26]
α_{FF6}	-0.17 [-2.40]	-0.09 [-1.45]	-0.05 [-0.74]	0.07 [1.37]	0.08 [1.63]	0.25 [2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EYR-sorted portfolios						
β_{MKT}	1.09 [60.90]	0.99 [66.87]	0.99 [62.24]	0.93 [68.43]	1.02 [86.73]	-0.07 [-2.89]
β_{SMB}	0.26 [10.22]	0.09 [4.30]	0.09 [3.80]	-0.06 [-3.19]	-0.11 [-6.31]	-0.36 [-11.12]
β_{HML}	0.09 [3.00]	0.19 [7.39]	0.08 [2.74]	-0.07 [-2.73]	-0.10 [-5.05]	-0.20 [-4.89]
β_{RMW}	-0.53 [-16.25]	-0.20 [-7.55]	-0.03 [-1.10]	0.13 [5.32]	0.23 [10.88]	0.76 [18.11]
β_{CMA}	-0.22 [-4.93]	-0.09 [-2.25]	0.10 [2.46]	0.02 [0.60]	0.09 [2.97]	0.31 [5.32]
β_{UMD}	-0.05 [-3.12]	-0.03 [-2.17]	-0.03 [-1.84]	-0.05 [-3.97]	0.04 [3.73]	0.09 [4.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1566	797	590	525	583	
me (\$10 ⁶)	1720	2163	2645	4552	5284	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EYR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.54 [3.32]	0.77 [5.16]	0.72 [6.07]	0.63 [5.33]	0.30 [3.26]	0.25 [2.68]
Quintile	NYSE	EW	0.38 [2.48]	0.46 [2.98]	0.34 [2.84]	0.24 [2.02]	0.02 [0.22]	-0.05 [-0.51]
Quintile	Name	VW	0.43 [2.30]	0.68 [3.90]	0.61 [4.35]	0.58 [4.03]	0.28 [2.23]	0.26 [2.06]
Quintile	Cap	VW	0.43 [3.00]	0.60 [4.33]	0.57 [4.92]	0.48 [4.17]	0.17 [1.88]	0.12 [1.29]
Decile	NYSE	VW	0.52 [2.85]	0.76 [4.39]	0.68 [4.67]	0.63 [4.29]	0.24 [1.96]	0.22 [1.78]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.51 [3.16]	0.75 [4.99]	0.69 [5.79]	0.63 [5.38]	0.34 [3.67]	0.30 [3.29]
Quintile	NYSE	EW	0.19 [1.23]	0.28 [1.73]	0.17 [1.36]	0.11 [0.92]		
Quintile	Name	VW	0.40 [2.13]	0.66 [3.76]	0.58 [4.13]	0.56 [3.95]	0.30 [2.35]	0.29 [2.23]
Quintile	Cap	VW	0.41 [2.87]	0.57 [4.10]	0.53 [4.57]	0.47 [4.14]	0.21 [2.24]	0.17 [1.83]
Decile	NYSE	VW	0.49 [2.67]	0.72 [4.12]	0.64 [4.35]	0.61 [4.12]	0.26 [2.08]	0.24 [1.94]

Table 3: Conditional sort on size and EYR

This table presents results for conditional double sorts on size and EYR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EYR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EYR and short stocks with low EYR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EYR Quintiles					EYR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.22 [0.49]	0.44 [1.22]	0.64 [2.18]	0.86 [3.25]	1.06 [3.41]	0.85 [3.20]	1.09 [4.18]	0.89 [4.22]	0.85 [3.94]	0.45 [2.38]	0.42 [2.20]
	(2)	0.33 [0.85]	0.56 [1.83]	0.77 [2.89]	0.93 [3.59]	1.03 [3.65]	0.70 [3.23]	0.94 [4.50]	0.77 [4.72]	0.75 [4.54]	0.33 [2.46]	0.32 [2.41]
	(3)	0.33 [0.96]	0.67 [2.37]	0.75 [3.06]	0.96 [3.98]	1.03 [3.88]	0.70 [3.80]	0.90 [4.97]	0.75 [5.26]	0.70 [4.83]	0.32 [2.68]	0.29 [2.44]
	(4)	0.64 [2.09]	0.68 [2.73]	0.82 [3.62]	0.88 [3.84]	0.92 [3.77]	0.28 [1.84]	0.45 [3.07]	0.35 [2.78]	0.31 [2.38]	0.01 [0.09]	-0.02 [-0.15]
	(5)	0.42 [1.67]	0.65 [2.89]	0.62 [3.07]	0.95 [4.69]	0.87 [4.20]	0.46 [3.15]	0.60 [4.25]	0.62 [4.57]	0.49 [3.70]	0.21 [1.84]	0.13 [1.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EYR Quintiles					EYR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	439	451	455	457	447	39	46	54	63	60	
	(2)	132	132	133	133	132	88	91	94	97	96	
	(3)	89	90	90	90	90	151	158	160	164	165	
	(4)	74	74	74	75	74	324	330	345	347	350	
(5)	66	66	66	66	66	2067	1875	2295	3256	3649		

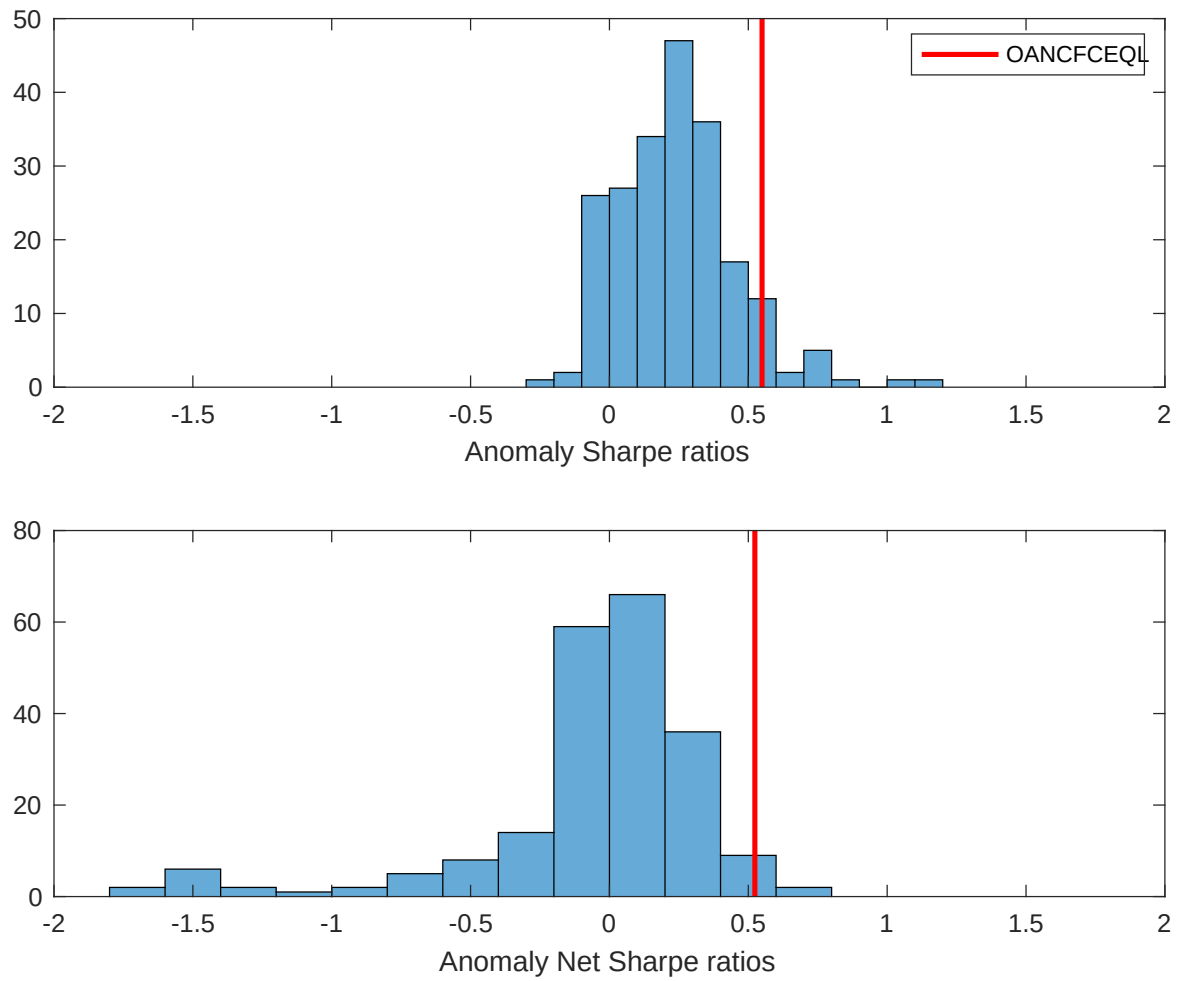


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EYR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

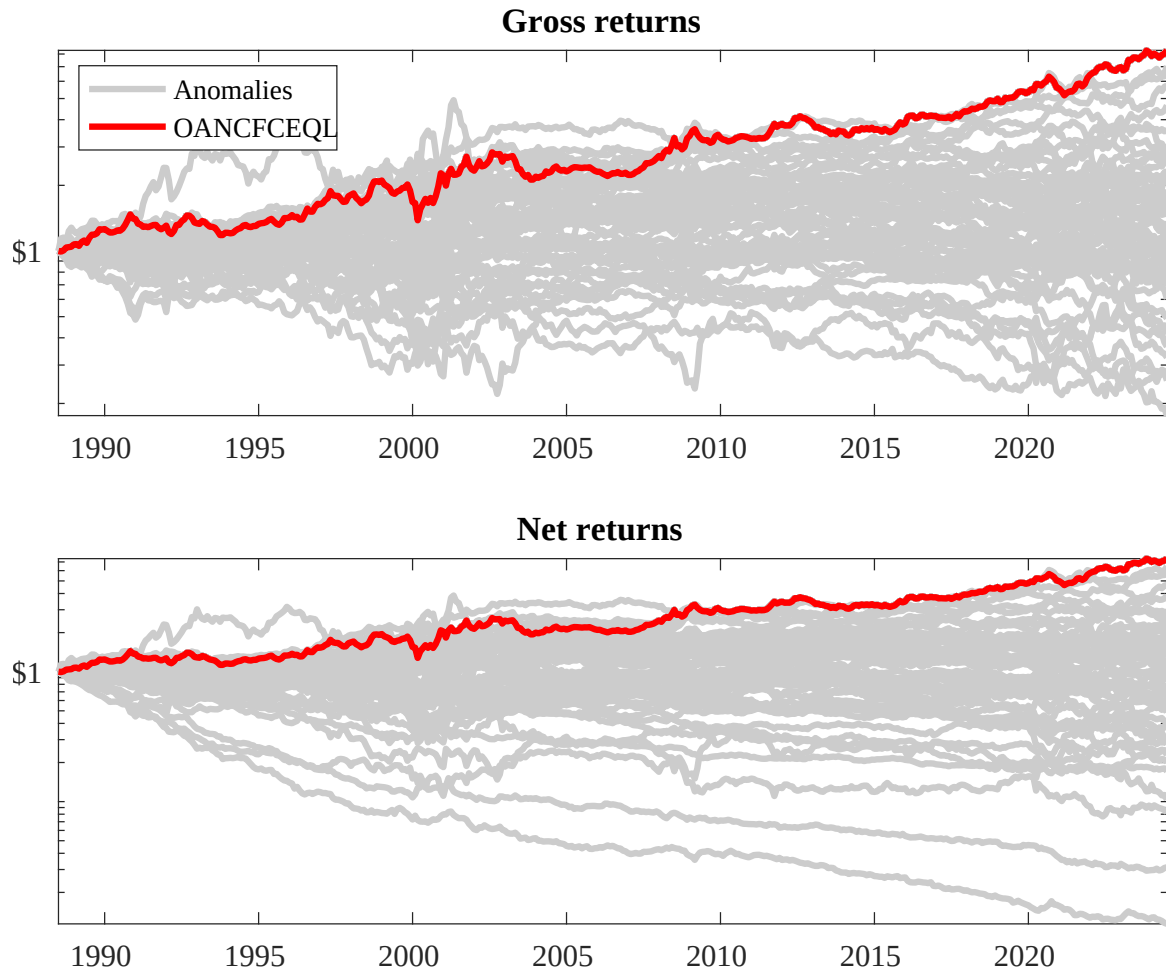
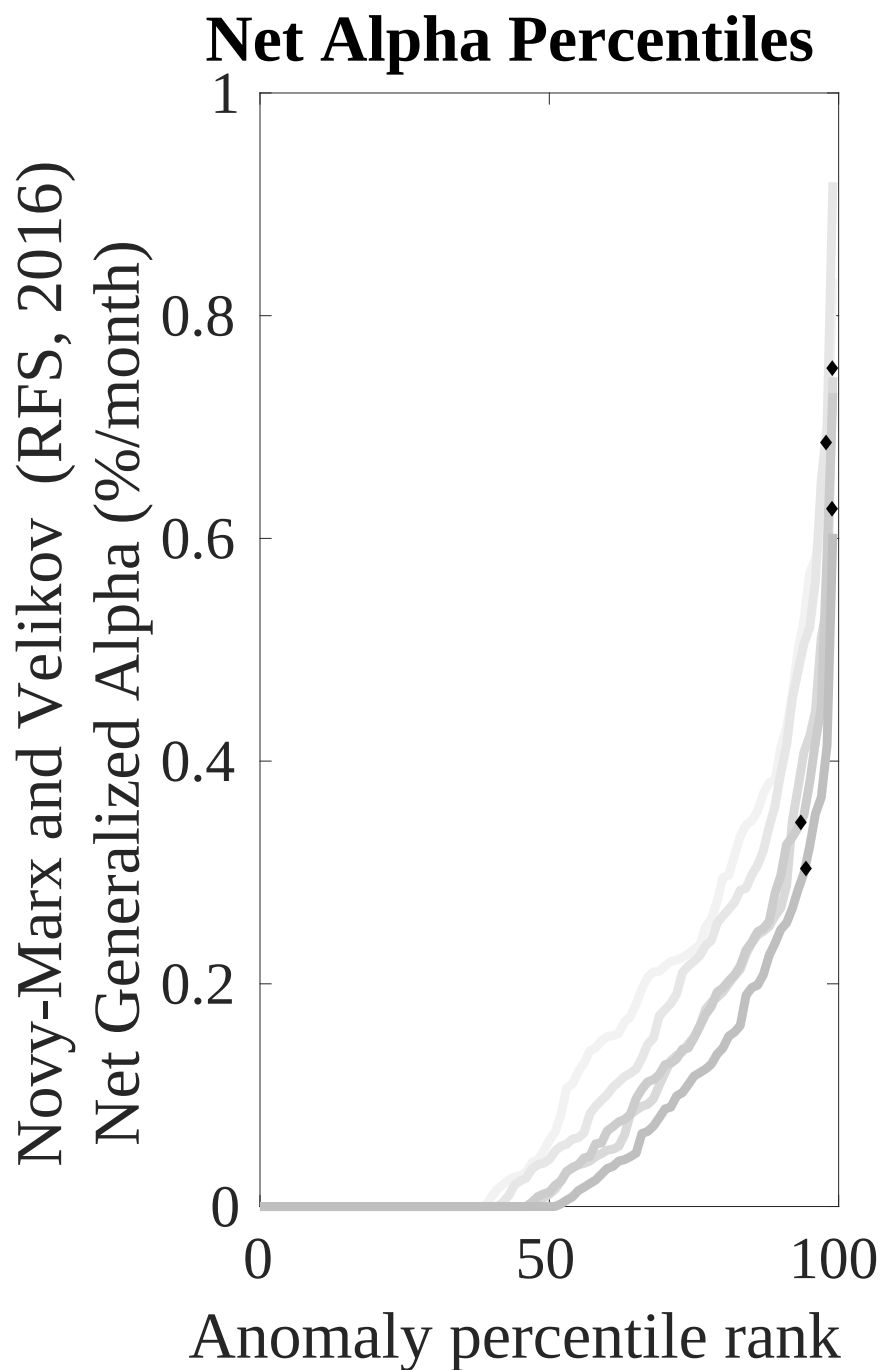
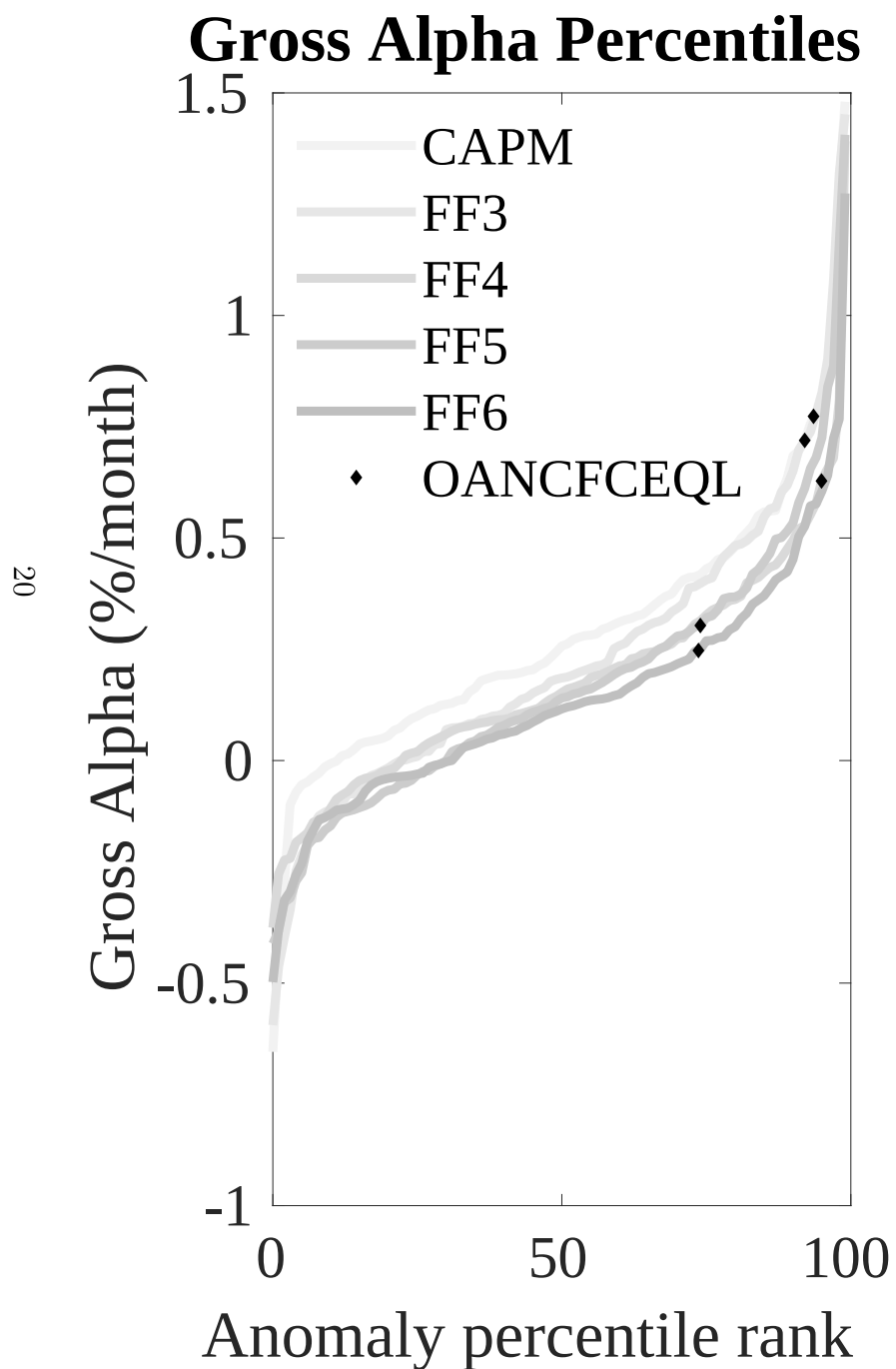


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EYR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



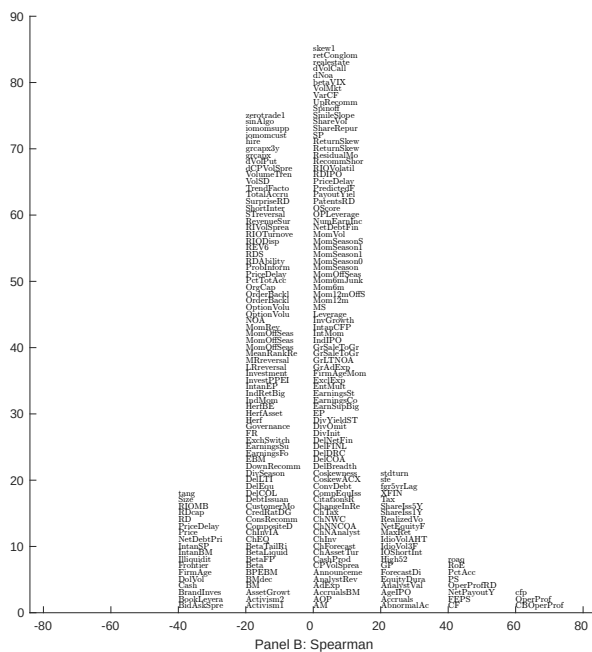
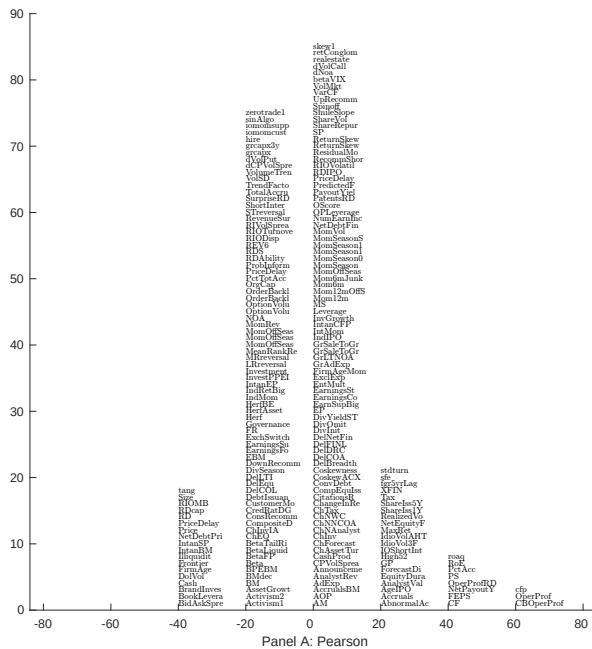


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EYR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

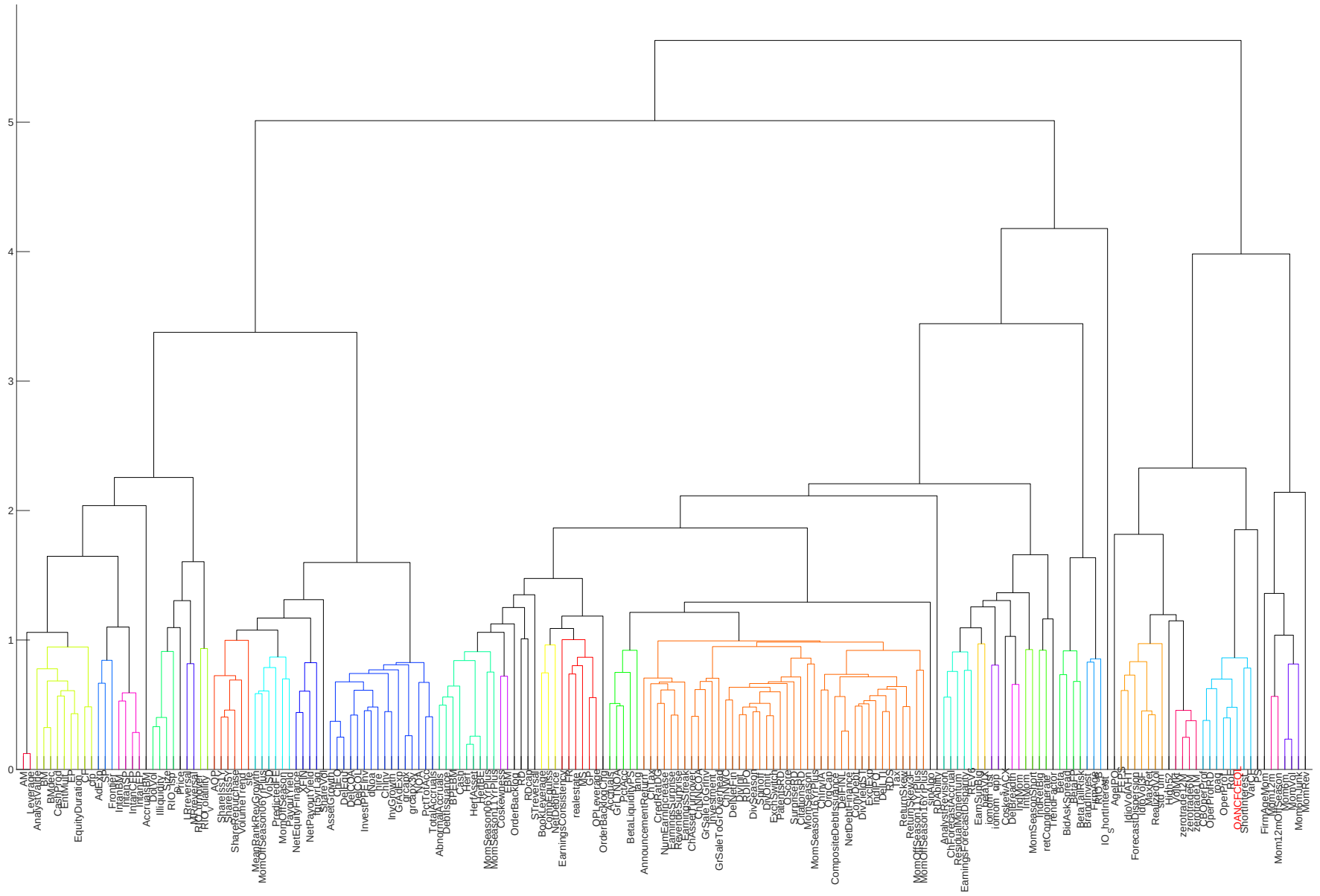


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

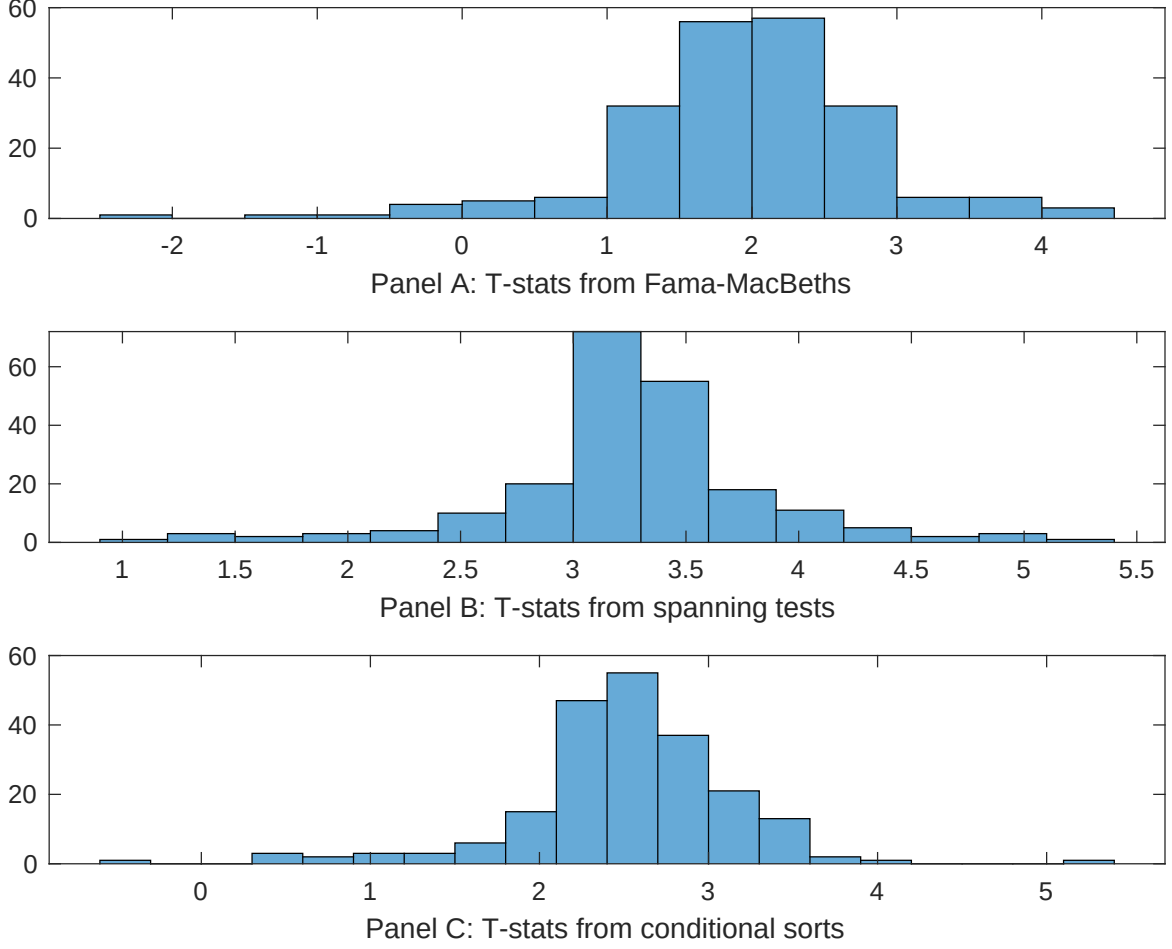


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EYR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.89 [3.02]	0.10 [3.49]	0.87 [2.54]	0.12 [4.27]	0.11 [4.78]	0.11 [4.00]	0.11 [5.08]
EYR	0.16 [2.62]	0.20 [2.80]	0.18 [2.96]	0.43 [0.06]	0.18 [3.71]	0.51 [0.97]	0.17 [2.62]
Anomaly 1	0.25 [2.79]						-0.21 [-0.28]
Anomaly 2		0.54 [0.01]					-0.14 [-2.61]
Anomaly 3			0.51 [0.74]				-0.14 [-0.48]
Anomaly 4				0.17 [5.67]			0.11 [4.38]
Anomaly 5					0.76 [1.25]		0.60 [0.75]
Anomaly 6						0.42 [2.69]	0.36 [3.16]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EYR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EYR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.51]	0.24 [2.59]	0.25 [2.70]	0.21 [2.28]	0.25 [2.70]	0.24 [2.57]	0.20 [2.32]
Anomaly 1	35.23 [8.57]						36.77 [7.53]
Anomaly 2		21.12 [3.80]					-3.97 [-0.63]
Anomaly 3			-2.83 [-0.80]				-17.49 [-4.23]
Anomaly 4				20.84 [4.68]			11.51 [2.37]
Anomaly 5					3.42 [1.15]		4.47 [1.34]
Anomaly 6						6.10 [1.48]	4.58 [1.10]
mkt	-2.52 [-1.15]	-3.49 [-1.45]	-7.02 [-2.87]	-3.39 [-1.44]	-4.97 [-1.89]	-5.52 [-2.31]	-2.61 [-1.07]
smb	-28.41 [-8.80]	-30.16 [-8.10]	-38.65 [-9.75]	-30.01 [-8.42]	-34.03 [-8.14]	-35.71 [-10.43]	-31.39 [-7.86]
hml	-14.09 [-3.72]	-17.30 [-4.29]	-18.62 [-4.55]	-16.93 [-4.24]	-19.97 [-4.83]	-17.05 [-3.99]	-9.97 [-2.42]
rmw	44.06 [8.28]	56.88 [8.95]	78.43 [13.66]	62.80 [12.86]	71.68 [13.65]	69.76 [12.40]	49.71 [7.48]
cma	23.79 [4.30]	34.20 [5.86]	32.59 [5.43]	18.42 [2.86]	30.44 [5.07]	31.37 [5.32]	18.95 [3.04]
umd	4.89 [2.49]	8.01 [3.92]	9.82 [4.48]	8.52 [4.23]	8.43 [3.89]	7.65 [3.31]	6.15 [2.78]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	76	72	72	73	72	72	77

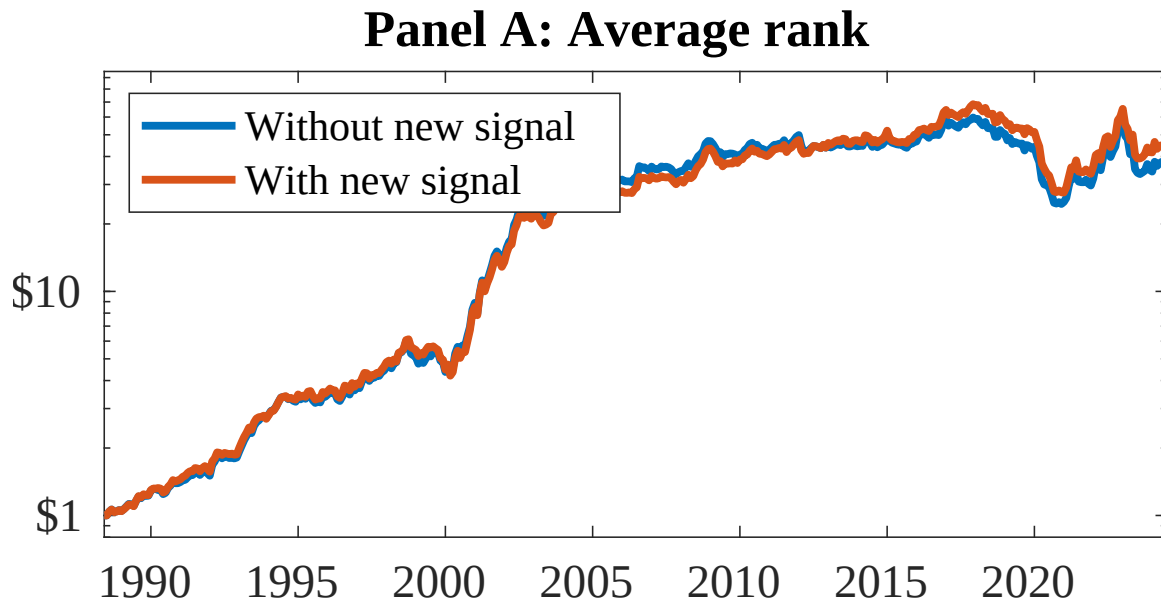


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as EYR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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